

Yinda Zhang

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SUMMARY

I am a Ph.D. student in Computer Science, with publications at top conferences in networking and database systems (including SIGCOMM, NSDI, KDD, SIGMOD, and VLDB). I also bring industry experience from Google, Meta, and ByteDance. My work spans the broad interdisciplinary areas of systems and networking.

EDUCATION

University of Pennsylvania

Advisor: Prof. Vincent Liu

University of Chicago

Advisor: Prof. Junchen Jiang

Peking University

Advisor: Prof. Tong Yang

Ph.D., Computer and Information Science

Sep. 2022 - Present

Pre-Doctoral M.S., Computer Science

Sep. 2020 - Dec. 2021

B.S., Computer Science and Technology

Sep. 2016 - Jul. 2020

INDUSTRY EXPERIENCE

Google | PhD Software Engineer Intern

Efficient Scheduling for ML Workloads on TPUs

- Enabled lifetime-aware scheduling for Cloud, reducing resource fragmentation
- Designed priority-aware bin-packing algorithms, maximizing utilization while minimizing preemptions

Sunnyvale, California

May. 2026 - Aug. 2026

Meta | PhD Software Engineer Intern

Scalability Improvements for Network Collection Pipeline

- Introduced sharding to improve fault tolerance and pipeline scalability
- Implemented local caching, reducing query load on the remote database

Menlo Park, California

May. 2025 - Aug. 2025

ByteDance | Applied Research Intern

Hardware Acceleration Architectures for Layer-4 Load Balancing

- Evaluated and compared hardware acceleration architectures for stateful L4 load balancing
- Performed detailed analysis of AMD Pensando DPU design trade-offs

Bellevue, Washington

May. 2024 - Aug. 2024

High-Performance Flow Logging via Matrix Sketching

- Designed a flow logging system that supports diverse query granularities
- Applied *matrix sketching* theory and implemented the system on the *Tofino* switch

May. 2024 - Aug. 2024

SELECTED RESEARCH PROJECTS

Data Transmission

Backward-Compatible Header Compression for Datacenter Networks

Sep. 2024 - Aug. 2026

- Designed a header compression mechanism that improves network efficiency for TCP/RDMA networks
- Implemented the solution using last-hop offload and MPLS encapsulation

Network Telemetry Transmission Without Affecting User Traffic

Nov. 2022 - Sep. 2024

- Developed a communication substrate for silently collecting network telemetry from switches
- Implemented the system on the *Tofino* switch and conducted large-scale simulations on *ns-3*

Data Approximation

Real-Time, Continuous Network Monitoring over Multiple Cores

Sep. 2021 - Sep. 2022

- Built a scalable software framework for sketch-based monitoring across multiple cores
- Applied the framework to nine sketch algorithms on three platforms: *CPU*, *DPDK*, and *XDP*

Near-optimal Global Summary for Aggregation in Wide Area Networks

Jun. 2021 - Mar. 2022

- Designed a near-optimal, adaptive, sampling scheme for WAN-scale data summarization
- Applied the algorithm to *federated learning*, *distributed state aggregation*, and *hierarchical aggregation*

SELECTED PUBLICATION

- Honey, I Shrunk the Headers With Flow.ZIP (*SIGCOMM 2026*)
- Enabling Silent Telemetry Data Transmission with InvisiFlow (*NSDI 2025*)
- Enabling Real-Time, Continuous Network Monitoring over Multiple Cores (*NSDI 2024*)
- A Near-optimal Global Summary for Aggregation in the Wide Area (*SIGMOD 2022*)
- High-Performance Sketch-based Measurement over Arbitrary Partial Key Query (*SIGCOMM 2021*)
- A Fast and Accurate Sketch on Persistence (*VLDB 2021*)
- A Framework using Time Zones for Data Stream Processing in Sliding Windows (*KDD 2020*)

MISC

Languages: C++/C, Python, P4, CUDA, Java, SQL

Systems & Tools: Network Simulator 3 (ns-3), programmable switch, DPDK, RDMA, eBPF

Fellowship: Jonathan M. Smith Fellowship (2022)